SOFTWARE REQUIREMENTS SPECIFICATION

HERBALANCE AI

AN AI-POWERED WOMEN'S HEALTH & WELLNESS COMPANION

An AI-Powered Women's Health and Wellness Platform

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification is to define the requirements of HERBALANCE AI – An AI-Powered Women's Health & Wellness Companion. The document specifies the intended functions, system behavior, interfaces, architecture, data requirements, security and privacy provisions, constraints, and evaluation criteria of the proposed system. It provides a common technical baseline for the project team, project guide, evaluators, testers, and future maintainers.

The SRS serves as a reference during system design, database development, AI/ML model development, implementation, integration testing, demonstration, and future maintenance of the application.

1.2 Scope

HERBALANCE AI is an AI-powered women's health and wellness platform designed to integrate multiple health and wellness services into a single application. The system provides personalized support through menstrual cycle prediction, skin type and skin tone analysis, cosmetic product recommendations, medicine safety verification, digital health record management, and personalized health insights.

The platform aims to improve accessibility, convenience, personalization, privacy, and informed decision-making in women's health and wellness. AI and machine learning techniques are applied to relevant user-provided information and images to generate predictions, classifications, recommendations, and personalized insights.

The system is intended to function as a supportive health and wellness platform and is not intended to replace professional medical diagnosis or treatment.

1.3 Objectives

The main objectives of HERBALANCE AI are to provide an integrated and user-friendly platform for women's health and wellness services. The system aims to predict menstrual cycles, analyze skin type and skin tone using AI, provide personalized cosmetic recommendations and medicine safety information, and securely manage digital health records. It also aims to generate personalized health insights, protect sensitive user information, and reduce the need for multiple independent applications.

1.4 Intended Audience

This document is intended for the project team, project guide, evaluators, testers, and future maintainers. It can also be used as a technical reference during database design, AI/ML development, system implementation, integration testing, project demonstration, and future system enhancement.

1.5 Definitions and Abbreviations

- AI – Artificial Intelligence
- ML – Machine Learning
- CNN – Convolutional Neural Network
- SRS – Software Requirements Specification
- UI – User Interface
- API – Application Programming Interface
- TFLite – TensorFlow Lite
- HERBALANCE AI – AI-powered Women's Health and Wellness Companion
- TLS – Transport Layer Security
- SSL – Secure Sockets Layer
- RMSE – Root Mean Square Error
- MAE – Mean Absolute Error
- F1-score – Harmonic mean of precision and recall

2. Overall Description

2.1 Product Perspective

HERBALANCE AI is an integrated AI-powered women's health and wellness platform. Instead of requiring users to depend on multiple applications for different wellness needs, the system provides several services through a unified application interface.

The platform consists of a user interface, application and data management components, AI/ML processing modules, secure data storage, and optional external data services. User-provided health information and images are processed by the appropriate modules to generate personalized predictions, classifications, recommendations, and insights.

2.2 Product Functions

The major functions of HERBALANCE AI include secure user registration and authentication, health profile management, menstrual cycle prediction, AI-based skin type and skin tone analysis, personalized cosmetic recommendations, medicine safety verification, and digital health record management. The system also generates personalized health insights, securely stores user information, and displays AI-generated results and recommendations through a user-friendly interface.

2.3 User Classes

The primary users of HERBALANCE AI are women seeking personalized health and wellness support. Users can create and manage their accounts, enter and update health information, track their menstrual cycles, upload images for skin analysis, receive cosmetic recommendations, check medicine-related safety information, manage digital health records, and view personalized health and wellness insights.

2.4 Operating Environment

HERBALANCE AI is intended to operate as a web-based platform accessible through modern web browsers. The operating environment includes a modern computer or compatible device, internet connectivity where required, a backend server for application processing, a secure database for data storage, AI/ML components for prediction and analysis, and external APIs or data sources where necessary.

2.5 Assumptions and Dependencies

The system assumes that users provide accurate health and wellness information, appropriate menstrual cycle data, and sufficiently clear skin images for analysis. It also assumes that the AI/ML models are trained using suitable datasets and that the required external APIs, data sources, network connectivity, and computational resources are available. The accuracy of predictions, classifications, and recommendations depends on the quality of the input data, datasets, AI/ML models, and external information sources.

3. System Analysis

3.1 Existing System

Existing women's health and wellness solutions commonly provide individual services through separate applications or manual methods. Menstrual tracking applications focus mainly on cycle tracking and prediction, while skincare applications may provide product recommendations or skin analysis. Medicine-related information is often obtained from separate websites, applications, or healthcare resources. Digital health records may also be stored separately or maintained manually.

These approaches generally focus on specific aspects of women's health rather than providing a single integrated platform.

3.2 Problems in Existing System

The major problems identified in existing approaches include the lack of an integrated women's health and wellness platform, requiring users to rely on multiple applications for different services. These systems may offer limited personalization and make it difficult for users to obtain suitable skincare recommendations and medicine-related safety information. In addition, health records may be scattered across different platforms, resulting in a lack of integration between wellness services and increased time and effort in managing personal health and wellness.

3.3 Problem Statement

Women often need to rely on multiple applications, websites, and manual methods to manage different aspects of their health and wellness. Existing solutions may provide individual services such as menstrual tracking, skincare recommendations, medicine information, or health record management, but these services are often disconnected.

Therefore, there is a need for an integrated AI-powered platform that provides menstrual cycle prediction, skin analysis, cosmetic recommendations, medicine safety verification, secure health record management, and personalized health insights through a single accessible and privacy-conscious system.

4. Proposed System

4.1 Proposed Solution

HERBALANCE AI is proposed as an integrated AI-powered women's health and wellness platform. The system combines multiple health and wellness services into a single application.

The system uses AI and ML techniques to analyze relevant user-provided information and images. The platform provides menstrual cycle prediction, AI-based skin analysis, personalized cosmetic recommendations, medicine safety verification, digital health record management, and personalized health insights.

The system is designed to provide convenient and personalized support while maintaining appropriate security and privacy of user information.

4.2 Advantages

The proposed system provides an integrated platform that offers multiple women's health and wellness services in one place. It supports menstrual cycle prediction, AI-based skin analysis, personalized cosmetic recommendations, medicine safety information, and secure digital health record management. The system provides centralized and convenient access to personalized health and wellness information, reduces dependency on multiple independent applications, and ensures the secure handling of sensitive user data. Its modular architecture also allows individual components to be improved and enhanced independently.

4.3 System Workflow

The general system workflow is:

User Registration/Login → Health Profile Management → User Data/Image Input → Appropriate AI/ML Module → Analysis/Prediction → Personalized Result/Recommendation → Secure Storage and Retrieval

4.4 Privacy and Safety Approach

Privacy and security are essential components of HERBALANCE AI due to the sensitive nature of personal and health-related information. The system shall protect user data, restrict access to authorized users, and securely handle health records and uploaded images. AI-generated

predictions and recommendations shall be presented only as supportive information and not as a replacement for professional medical advice.

4.5 Personalized and Adaptive Features

The system uses available user information, such as health profiles, menstrual cycle data, skin characteristics, preferences, medicine information, and health records, to generate personalized recommendations and insights relevant to individual users.

5. System Architecture

5.1 Architecture Components

The system consists of several major architectural components. The User Interface Layer provides access to system features, collects user inputs, and displays results. The Application Layer manages user requests, authentication, and communication between different components. The AI/ML Processing Layer performs menstrual cycle prediction, skin analysis, and personalized recommendations. The Data Management Layer securely stores user profiles, health records, and analysis results. The External Service Layer connects the system with required APIs and external data sources, while the Security Layer ensures authentication, authorization, and the secure handling of sensitive information.

5.2 AI/ML Processing Pipeline

The AI/ML processing pipeline varies according to the selected module. For menstrual prediction, the system processes menstrual data and uses a prediction model to generate the expected cycle information. For skin analysis, uploaded skin images are preprocessed and analyzed using an AI/ML model to identify skin type and skin tone. Similarly, user profile information and analysis results are processed to generate personalized recommendations, which are then displayed through the user interface.

5.3 Application and Data Flow

The user enters or uploads relevant information through the user interface. The application layer validates and processes the input before passing it to the appropriate module.

The AI/ML module performs the required analysis or prediction. The resulting output is returned to the application layer and displayed to the user. Relevant information may be securely

stored in the database for future authorized retrieval.

5.4 External Service Integration

External data sources or APIs may be integrated where required for:

- Medicine information.
- Cosmetic product information.
- Health-related information.
- Other supported external services.

The exact external APIs and services used in the final implementation shall be documented by the project team.

6. Detailed Module Description

6.1 User Registration and Profile Management

This module allows users to create and manage their accounts. Users can register using the required information and authenticate securely.

The module also allows users to enter and update relevant health and wellness information. The profile information can be used by other modules to provide personalized services.

6.2 Menstrual Cycle Prediction

The Menstrual Cycle Prediction module uses user-provided menstrual cycle information to predict upcoming cycle patterns.The module analyzes available cycle information and generates predicted cycle-related information. The prediction can help users track their menstrual cycle and plan accordingly.The quality of prediction depends on the accuracy and completeness of the menstrual information provided by the user and the performance of the selected prediction model.

6.3 Skin Type and Skin Tone Analysis

This module uses AI-based image analysis to identify the user's skin type and skin tone.The user uploads a suitable image through the application. The image is processed by the AI/ML

component, which generates the corresponding skin analysis result.The resulting skin characteristics can be passed to the cosmetic recommendation module.

6.4 Cosmetic Product Recommendation

This module provides personalized cosmetic product recommendations based on the user's skin characteristics and preferences. The recommendation process may use:

- Skin type.
- Skin tone.
- User preferences.
- Available product information.

The module aims to help users identify cosmetic products that may be more suitable for their individual characteristics.

6.5 Medicine Safety Verification

The Medicine Safety Verification module provides medicine-related safety information based on the medicine details provided by the user. The module can identify relevant safety information and potential concerns associated with the provided medicine information. The output is intended to improve awareness and support informed decision-making. It shall not be treated as a substitute for professional medical advice.

6.6 Digital Health Record Management

This module provides secure storage and management of digital health records.Users can store, view, and manage relevant health information through the application. Access to these records shall be restricted to authenticated and authorized users.The module aims to provide centralized access to important health-related information while maintaining appropriate security and privacy.

6.7 Personalized Health Insights

This module analyzes available user information to generate personalized health and wellness insights.Relevant information from the user's profile and supported modules can be used to

provide meaningful wellness-related information.The generated insights are intended to support awareness and preventive health management rather than provide medical diagnosis.

7. Functional Requirements

ID	Requirement
FR-01	The system shall allow users to register and authenticate securely.
FR-02	The system shall provide a personalized dashboard for authenticated users.
FR-03	The system shall allow users to create, view, and update their health and wellness profile.
FR-04	The system shall allow users to enter menstrual cycle information.
FR-05	The Menstrual Cycle Prediction module shall analyze user-provided menstrual data and provide predicted cycle information.
FR-06	The system shall allow users to upload images for skin analysis.
FR-07	The Skin Analysis module shall identify the user's skin type and skin tone using AI-based image analysis.
FR-08	The Cosmetic Recommendation module shall provide personalized product recommendations based on available skin characteristics and user preferences.
FR-09	The system shall allow users to enter medicine information for safety verification.
FR-10	The Medicine Safety Verification module shall provide relevant safety information based on the provided medicine information.
FR-11	The system shall allow users to securely store digital health records.
FR-12	The system shall allow authenticated users to view and manage their authorized health records.
FR-13	The system shall generate personalized health and wellness insights using available user information.
FR-14	The system shall display predictions, classifications, recommendations, and health insights through the user interface.

FR-15	The system shall securely store relevant user information and module results for authorized retrieval.
FR-16	The system shall restrict personal and health information to authenticated and authorized users.
FR-17	The system shall provide a user-friendly interface for accessing the different modules.
FR-18	The system shall handle invalid or incomplete user inputs appropriately.
FR-19	The system shall securely handle uploaded images used for skin analysis.
FR-20	The system shall maintain appropriate records of supported health and wellness information.

7.1 Functional Behavior Summary

The functional requirements define a system in which the user registers and provides relevant health and wellness information. The system processes the information using the appropriate modules and generates predictions, classifications, recommendations, or insights.

The major processing flow consists of:

User Input → Validation → AI/ML or Application Processing → Result Generation → Display → Secure Storage where applicable

The system shall maintain appropriate access control throughout this process to protect user information.

8. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	User health information shall be protected using secure authentication, access control, and appropriate data security mechanisms.
NFR-02	Privacy	Personal and health-related information shall not be exposed to unauthorized users.

NFR-03	Performance	The system shall provide reasonably fast responses for normal user requests, predictions, analysis, and recommendations.
NFR-04	Usability	The application shall provide a simple, clear, and user-friendly interface.
NFR-05	Data Integrity	User health records and stored information shall be maintained accurately and consistently.
NFR-06	Scalability	The system shall be capable of supporting an increasing number of users and stored health-related information without significant degradation under expected operating conditions.
NFR-07	Reliability	The system shall provide consistent operation under normal operating conditions.
NFR-08	Maintainability	The system shall use a modular design so that individual components can be updated or improved independently.
NFR-09	Accessibility	The application shall be accessible across supported devices and modern web browsers and provide a consistent user experience.
NFR-10	Explainability	AI-generated results and recommendations should be presented in a manner that is understandable to users and clearly distinguished from professional medical advice.

8.1 Quality Goals

The system shall prioritize the security and privacy of sensitive user information, reliable operation, reasonable AI/ML performance, data integrity, and ease of use. It shall also focus on maintainability, scalability, accessibility, and providing understandable system outputs.

9. External Interface Requirements

9.1 User Interface

The application shall provide a clear and user-friendly interface for user registration and login, health profile management, menstrual cycle data entry and prediction, skin image upload and

analysis, cosmetic recommendations, medicine safety verification, digital health record management, and personalized health insights. The interface shall clearly display system results and provide suitable navigation between different modules.

9.2 Application Interface

The application interface shall enable communication between the user interface, application logic, database, AI/ML modules, and external services. It shall support the efficient transfer of required input data and processing results between these system components.

9.3 AI/ML Interface

The AI/ML modules shall receive relevant input data from the application and generate predictions, classifications, or personalized recommendations. For example, menstrual data can be used for cycle prediction, skin images for skin type and skin tone analysis, and user characteristics for personalized recommendations. The generated results shall be appropriately processed and displayed through the application interface.

9.4 External Data Interfaces

External data sources or APIs may be used to obtain medicine information, cosmetic product information, and other relevant health and wellness data. The specific external data sources and APIs used in the final implementation shall be documented by the project team.

10. Data Requirements

The system shall process and maintain structured information, including user account details, health profiles, menstrual cycle information, skin images and analysis results, cosmetic recommendations, medicine safety information, digital health records, and personalized health insights. User-provided health information and images shall be protected using appropriate privacy and security controls. AI/ML datasets used during model development shall be properly documented, and user health information shall be kept separate from model-development datasets wherever applicable to protect personal information.

10.1 Primary Data Categories

The principal data categories include user data, health and wellness information, menstrual cycle details, skin images and analysis results, cosmetic recommendations, medicine information and safety results, digital health records, and personalized health insights.

11. Database Requirements

Entity	Key Fields	Purpose
User	user_id, name, email, password_hash	User authentication and account management
HealthProfile	profile_id, user_id, age, health_details	Stores user health and wellness information
MenstrualRecord	menstrual_id, user_id, cycle_start_date, cycle_length, symptoms	Menstrual cycle tracking and prediction
SkinAnalysis	analysis_id, user_id, image_reference, skin_type, skin_tone, analyzed_at	Stores AI-based skin analysis results
CosmeticRecommendation	recommendation_id, user_id, skin_type, skin_tone, products, created_at	Stores personalized cosmetic recommendations
MedicineCheck	check_id, user_id, medicine_name, safety_result, checked_at	Stores medicine safety verification results
HealthRecord	health_record_id, user_id, record_type, record_data, created_at	Digital health record management
HealthInsight	insight_id, user_id, insight_type, content, created_at	Stores personalized health and wellness insights

11.1 Data Handling

The database shall retain information required for user authentication, health profile management, menstrual cycle tracking, skin analysis, cosmetic recommendations, medicine verification, digital health records, and personalized health insights. Sensitive information shall be securely stored and accessible only to authorized users, while appropriate relationships shall be maintained between user accounts and their corresponding health and wellness information.

12. Use Case Analysis

ID	Use Case	Description
UC-01	User Registration and Login	User creates an account and authenticates securely.
UC-02	Health Profile Management	User enters, views, and updates relevant health and wellness information.
UC-03	Menstrual Cycle Prediction	User provides menstrual cycle data and receives predicted cycle information.
UC-04	Skin Analysis	User uploads an image and receives AI-based skin type and skin tone analysis.
UC-05	Cosmetic Recommendation	User receives personalized cosmetic product recommendations based on available skin characteristics and preferences.
UC-06	Medicine Safety Verification	User provides medicine information and receives relevant safety information.
UC-07	Health Record Management	User securely stores, views, and manages digital health records.
UC-08	Personalized Health Insights	System analyzes available user information and provides personalized health and wellness insights.
UC-09	View Results and Recommendations	User views predictions, analysis results, recommendations, and insights.

UC-10	Secure Data Access	System allows only authenticated and authorized users to access personal health information and records.

13. Hardware and Software Requirements

13.1 Hardware Requirements

The system requires a modern computer with sufficient processing power, memory, and storage for application development and AI/ML processing. A camera or mobile/web camera is required for capturing images for skin analysis, while adequate storage is needed for application files, datasets, models, and health-related data. Internet connectivity is also required for accessing server-side services and external APIs. The exact hardware configuration may vary depending on the selected AI/ML models and implementation approach.

13.2 Software Requirements

The system requires web-based frontend technologies, a server-side backend environment, and a secure database for managing application and health-related information. Python and other required web development technologies will be used for implementation, while TensorFlow, TensorFlow Lite, and relevant machine learning libraries will support AI/ML processing. The system may also use image processing libraries, modern web browsers, suitable development environments, and external APIs for medicine, cosmetic, and other health-related information.

14. Security and Privacy Requirements

The system shall implement appropriate security and privacy measures to protect sensitive personal and health-related information. User passwords shall be securely hashed, and authentication and authorization mechanisms shall prevent unauthorized access to user data and health records. Uploaded skin images and database information shall be securely handled, while secure communication protocols shall be used where applicable to protect data during transmission and processing.

14.1 Data Privacy

Privacy shall be maintained throughout the collection, processing, storage, and retrieval of user information.The system shall collect only information necessary for the supported functionality. Sensitive information shall be stored securely and access shall be limited to authorized users.Uploaded images used for skin analysis shall be handled securely and should not be exposed beyond the requirements of the system.

14.2 Secure Health Record Access

Digital health records shall be protected using authentication and authorization mechanisms.Only the authenticated user or an explicitly authorized role, if implemented, shall be permitted to access the corresponding health records.

15. System Constraints

The system is subject to constraints related to the quality and availability of datasets, user-provided information, image quality, external APIs, network connectivity, and computational resources. The accuracy of AI/ML predictions, skin analysis, and recommendations may vary depending on these factors. The project must also remain feasible within the available time and resources of a B.Tech final-year project. Furthermore, AI-generated predictions, recommendations, and insights shall be considered supportive information and not a substitute for professional medical diagnosis or treatment.

16. Testing and Evaluation Requirements

16.1 Functional Testing

Functional testing shall verify that each system feature operates according to the specified functional requirements. Testing shall include user registration and authentication, profile management, menstrual data entry and prediction, image upload and skin analysis, recommendation generation, medicine verification, digital health record management, personalized insight generation, and prevention of unauthorized access.

16.2 AI/ML Performance Evaluation

AI/ML modules shall be evaluated using appropriate performance metrics based on their specific tasks. Menstrual cycle prediction may be evaluated using error-based metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Square Error (RMSE). Skin classification and analysis may be evaluated using metrics such as accuracy, precision, recall, F1-score, sensitivity, specificity, and balanced accuracy. Other recommendation or prediction modules may use suitable metrics depending on the final implementation and the type of model used.

16.3 Usability, Security and Reliability Testing

The complete system shall be evaluated based on user interface clarity, ease of navigation, response time, data integrity, authentication, privacy protection, secure data handling, reliability, error handling, recovery from failures, and compatibility across supported browsers and devices. The system should also provide understandable outputs and clearly communicate that AI-generated results are supportive information and not a substitute for professional medical advice.

17. Future Scope

The future development of HERBALANCE AI can include:

17.1 Wearable Device Integration

Wearable devices such as smartwatches and fitness bands can be integrated to collect relevant health and wellness information continuously.

17.2 Multilingual and Voice-Based Interaction

The system can be extended with multilingual support and voice-based interaction to improve accessibility and usability.

17.3 Telemedicine Support

Future versions can provide integration with healthcare professionals and telemedicine platforms, allowing users to seek professional advice when required.

17.4 Advanced Personalized Health Insights

More advanced AI models can analyze long-term user information to generate improved personalized wellness insights while maintaining appropriate privacy controls.

17.5 Digital Health Record Integration

The system can be extended to support interoperability with other digital health record systems and healthcare services where technically and legally appropriate.

18. Conclusion

HERBALANCE AI is designed as an integrated AI-powered women's health and wellness platform that combines menstrual cycle prediction, skin type and skin tone analysis, cosmetic product recommendations, medicine safety verification, digital health record management, and personalized health insights.

By applying AI and machine learning techniques to relevant user-provided data and images, the system aims to provide personalized, accessible, convenient, and privacy-conscious health and wellness support through a unified platform.

The proposed system reduces the dependency on multiple independent applications and provides users with centralized access to several women's health and wellness services. Secure authentication, access control, and appropriate data protection mechanisms are incorporated to safeguard sensitive personal and health-related information.

This Software Requirements Specification establishes the baseline requirements for the design, implementation, integration, testing, and evaluation of HERBALANCE AI as a B.Tech final-year project. The modular design also provides a foundation for future enhancements such as mobile applications, wearable integration, multilingual interaction, advanced AI models, telemedicine support, and expanded health services.

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